Graph based EEG Analysis

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***Abstract*—Electroencephalography (EEG) signals offer a non-invasive window into brain activity but present significant challenges due to their high dimensionality, noise, and non-stationarity. This study explores the potential of graph-based models to enhance EEG-based seizure detection by explicitly capturing spatial or functional relationships between electrodes, complementing traditional temporal modeling. We evaluate six graph-aware architectures including GNNs, spatio-temporal models (STGCN, DCRNN), and hybrid approaches (LSTM-GNN, GNN-LSTM, S4-GNN) against three temporal baselines (Bi-LSTM, ResNet-LSTM, S4) using a subset of the Temple University Hospital EEG Seizure Corpus. The dataset comprises 12-second, 19-channel recordings from 97 patients, with rigorous patient-separated splits to ensure generalizability. Models are optimized for macro F1 to address class imbalance (20% seizure prevalence). Our results demonstrate that integrating spatial and temporal features significantly improves performance: the S4-GCN hybrid achieves a test F1 of 0.838, outperforming pure temporal baselines achieving a test F1 of 0.814 for S4 and standalone GNNs achieving a test F1 of 0.569. Frequency-domain preprocessing generally enhanced detection, though some graph variants STGCN, S4-GNN and a non-graph variant Resnet-LSTM performed optimally with time-domain data. These findings underscore the importance of combining long-range temporal modeling with explicit spatial structure, offering a robust framework for EEG-based seizure detection. The success of hybrid models like S4-GCN highlights the potential of graph representations to advance the accuracy of clinical EEG analysis.**

I. Introduction

Epilepsy is a neurological disorder that affects around 50 million people worldwide [1], characterized by recurrent seizures caused by abnormal, excessive, or synchronous neuronal activity in the brain. A widely used non-invasive method for monitoring brain activity is Electroencephalography (EEG) [2], which records electrical signals from electrodes placed on the scalp [3]. These recordings produce multichannel time-series data reflecting neural activity across different brain regions. However, EEG signals are often noisy, high-dimensional, and non-stationary, making them challenging to analyze. Traditional approaches rely on signal processing techniques or time-series models like RNNs and transformers, which often neglect the spatial relationships between electrodes which can be an important factor for understanding brain connectivity.

Improving seizure detection through more accurate modeling of EEG data could significantly impact patient care. It is estimated that 70% of people with epilepsy could live seizure-free if properly diagnosed and treated. For those who are resistant to treatment, early detection and even prediction of seizures can enhance quality of life by enabling timely interventions. Graph-based models, which represent EEG electrodes as nodes and their functional or spatial relationships as edges, offer a promising direction for capturing these interdependencies. By comparing such models with traditional approaches, this project aims to understand whether incorporating spatial structure can improve the performance and interpretability of EEG-based seizure detection systems.

II. Methods

A subset of the Temple University Hospital EEG Seizure Corpus (TUSZ) [4], which is one of the largest public EEG seizure databases, was provided for this project. This dataset consists of 12 second non-overlapping EEG recordings sampled at 250Hz from 19 electrodes placed according to the international 10–20 system (Figure 1). Recordings from 97 patients are used for training and 25 for testing, with each patient having multiple EEG sessions. Each session captures voltage fluctuations in millivolts corresponding to brain activity. For the training set, each window is labeled to indicate whether it contains regular brain activity or seizure activity. Additionally, pairwise distances between electrodes are provided to support the construction of spatial graphs. The dataset contains approximately 13,000 segments in total. Notably, seizure events constitute only about 20% of the dataset, making this a significantly imbalanced classification problem.

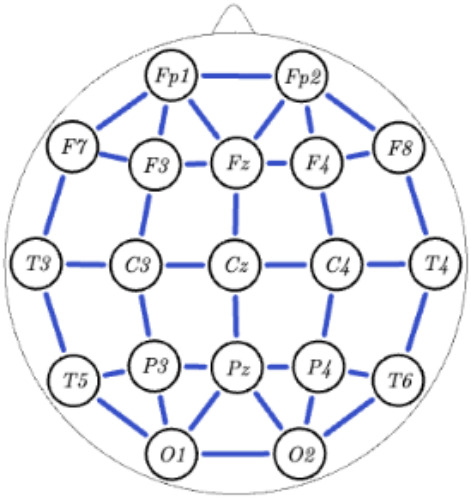


Fig. 1. 10-20 EEG system

A. Data Preprocessing

Each sample in the dataset consists of 19 time series, each with 3000 time steps, corresponding to 12 second EEG recordings at 250Hz. Because seizures are associated with brain electrical activity in certain frequency bands [9], the data was transformed into the frequency domain using the Fast Fourier Transform (FFT), which often reveals more discriminative patterns for seizure detection. Since relevant EEG activity typically lies within the 0.5–30Hz range, a frequency filter was applied to retain only the meaningful components and suppress noise outside this band. When working in the time domain, an equivalent band-pass filter was employed to achieve the same effect. For validation, data from nine unique patients maintaining a balance between seizure and non seizure samples were selected, while recordings from the remaining eighty-eight patients were used for training. This approach was adopted to ensure robust validation, as using signals from the same patients in both training and validation could lead to data leakage. By separating patients, it was ensured that the model's performance reflects its ability to generalize to unseen individuals. Next subsection goes through different methods explored to solve this problem.

B. Baseline Methods

1) BiLSTM: BiLSTMs [17] are a powerful variant of recurrent neural networks (RNNs) that process sequential data in both forward and backward directions, enabling them to capture complex temporal dependencies in EEG signals more effectively than standard LSTMs [18]. While LSTMs only consider past context, BiLSTMs model long-range interactions by integrating information from both preceding and subsequent time steps. This bidirectional capability can be critical for seizure prediction, where pathological patterns often exhibit subtle, context-dependent evolution that requires full temporal context for robust detection. In our experiments, three layer standard BiLSTMs followed by a classification layer were used as a simple baseline, as they were found to outperform standard LSTMs in EEG seizure detection.

2) Resnet-LSTM: Alshaya et. al in [10] proposed a deep learning architecture that leverages a 1D Residual Network (ResNet) to hierarchically extract multi-scale temporal features directly from raw EEG time-domain signals, thereby capturing local patterns indicative of seizure activity. Then these learned temporal features are fed into a LSTM network, which models the long-range sequential dependencies and temporal evolution characteristic of different seizure types, followed by two fully connected layers to predict the seizure type. Additionally, the authors employed the SMOTE [11] technique to synthetically balance the dataset and address class imbalance; in this work, positional weight is also tested. In this adaptation, the ResNet-LSTM model was implemented following the architecture proposed in [10], with an additional fully connected layer using a ReLU activation, followed by a final fully connected layer to produce a binary prediction output.

3) S4: Structured State Space Sequence (S4) [14] models are designed to efficiently capture long-range dependencies in sequential data using state-space representations. This makes them particularly well-suited for EEG seizure detection, where important patterns often unfold over extended periods of time. In this baseline model, a single S4 layer is applied to each EEG channel. The output is mean pooled over time for each channel, then averaged across channels to produce a global representation, and finally passed to a classification head to predict seizure activity.

C. Graph based Methods

Unless specified otherwise, Graph used in all methods below is constructed using the standard 10-20 system where edge weights are derived from the inverse of the pairwise given physical distances between EEG electrodes, thereby emphasizing stronger connections between spatially proximal nodes.

1) GNN: In this approach, frequency-domain representations of the EEG signals are used as node features for a Graph Neural Network (GNN) [8] particularly Graph Attention Networks (GATs) [16], since temporal dynamics are not explicitly modeled. The GNN captures spatial dependencies among electrodes, and its output is passed through a classification layer for final prediction.

2) LSTM-GNN: This approach combines two complementary models: a Bidirectional Long Short Term Memory (Bi-LSTM) model to capture temporal dynamics within each EEG channel, and a GNN (GCNs [15], GATs or Graph Transformers [19]) to model spatial dependencies across electrodes. First, the data from each node is processed by the LSTM to produce temporal embeddings. These embeddings are then passed through the GNN, which updates each node's representation by aggregating information from its neighboring nodes. Finally, the graph-level representation is obtained through pooling and fed into a classification layer to predict seizure activity. This architecture was implemented in three variants: Bi-LSTM + GCN, Bi-LSTM + GAT, and Bi-LSTM + Graph Transformer. All variants shared a common structure consisting of a three-layer Bi-LSTM followed by a three-layer GNN with residual connections.

3) GNN-LSTM: This spatio-temporal approach reverses the LSTM-GNN sequence by first utilizing the GNN to capture spatial correlations between electrodes. The data at each time step are processed by separate GNNs to extract spatially-aware features for each node. These features are then concatenated over time to form sequences that are inputted into the LSTM, which models the temporal dependencies across these spatial embeddings. The last timestep output is used to predict seizure activity. Four variants were explored: 1 Layer GCN+LSTM, 2 Layers GCN+LSTM, 1 Layer GAT+LSTM, 2 Layers GAT+LSTM. All variants used 3 Layers of LSTM.

4) STGCN: In this approach from Yu et al. [5], originally proposed for traffic forecasting, the STGCN architecture is adapted to EEG-based seizure detection. Although initially developed for time series prediction tasks, the model's ability to jointly capture spatial and temporal dependencies made it suitable for EEG analysis, where signals from spatially

distributed electrodes evolve over time. Graph convolutions are used to capture spatial relationships between electrodes, leveraging a first-order approximation of spectral graph filters for computational efficiency. Temporal dependencies are modeled through gated 1D convolutions applied independently along the time axis for each node, allowing local temporal features to be learned effectively. To adapt the model for classification, two variants where implemented. The first variant applied a simple Adaptive Average Pooling layer prior to a sequence of two fully connected layers with ReLU activations, followed by a final output layer for binary classification, which replaces the forecasting head. The second variant replaced the average pooling with an attention-based pooling mechanism, while retaining the same fully connected architecture.

5) S4-GNN: In this method, to account for long-range temporal dependencies in EEG signals, a hybrid architecture combining the S4 model with a graph-based component is used. The S4-GNN implementation first applies a single S4 layer to each EEG channel to extract temporal representations. A GNN (GCN or GAT) is then used to model spatial relationships between electrodes. The resulting embeddings are pooled and passed to a classifier for seizure prediction.

6) DCRNN: Diffusion Convolutional Recurrent Neural Network [13], originally developed for traffic forecasting [12], capture the spatio-temporal dependencies by modeling the spatial relationships among electrodes as a diffusion process (modeled by a bidirectional random walk) on the electrode graph. The graph can be distance based to capture EEG sensor geometry or correlation based to capture dynamic connectivity in the brain. Temporal dynamics are captured by Gated Recurrent Units (GRUs), with matrix multiplications replaced by diffusion convolutions. Hence, the architecture consists of multiple stacked diffusion convolutional GRU layers (DCGRUs), followed by a fully connected output layer for seizure detection.

III. Results

This section first presents the results of each individual model along with the details of their fine-tuning, followed by a series of comparative studies analyzing the impact of various design choices.

All models were trained using a hold-out validation set, as previously described using macro F1 as the validation metric. Checkpoints were saved whenever the validation score improved, and the best-performing model was subsequently used for testing. This was done to avoid overfitting to the training data and to ensure that model selection was based on generalization performance rather than memorization. All models were also trained using a weighted binary cross-entropy loss to address class imbalance, ensuring more balanced learning despite the disparity in seizure and non-seizure samples.

A. Baseline Methods

1) Bi-LSTMs: A three Layer Bi-LSTM model was trained for 200 epochs with learning rate of 5×10^{-5} which yielded 0.729 test F1. As this model served as a baseline, extensive hyperparameter tuning was not performed. Nonetheless, it demonstrated strong performance out of the box, highlighting the effectiveness of temporal modeling for EEG-based seizure detection.

2) Resnet-LSTM: The training was conducted for 200 epochs using a learning rate of 5×10^{-5}, a weight decay of 1×10^{-5}, and gradient clipping with a maximum gradient norm of 5.0. On the robust validation set, the model achieved an F1 score of 0.79. Interestingly, the model's validation F1 score plateaued at around 0.70–0.72 after the initial 10 epochs, where it had already reached its peak performance. But when trained without robust validation (i.e., using 95% of the data for training), the model achieved a test score of 0.81 on test set, with the best model found around epoch 150. This model achieved an F1 score of 0.83 on the remaining 5% of the data used for testing. However, since patient-level separation was not enforced in this split, the possibility of data leakage exists; therefore, the Kaggle test set evaluation score is a more reliable metric to consider here.

Furthermore, training the model on frequency-domain representations led to a decline in performance, with a best F1 score of only 0.72, and also the model in this case overfit more rapidly and failed to generalize well.

3) S4: This method uses a S4 layer to process frequency data from each channel independently, then averages them across both time and electrodes before classification. The model was trained for 300 epochs with a learning rate 1×10^{-3} and yielded a 0.814 macro F1 on the test data. This result highlights the effectiveness of state-space models for EEG-based seizure detection.

B. Graph based Methods

1) GNN: A three-layer GAT model with residual connections was trained using frequency-domain features obtained from FFT as node inputs. The model was optimized with the Adam optimizer for 300 epochs, using a learning rate of 5×10^{-5} and a weight decay of 1×10^{-5}. It achieved a test F1 score of 0.57, while the training F1 score approached 1. However, the validation F1 plateaued around 0.65, indicating signs of overfitting. Further tuning of the hyperparameters did not lead to any noticeable improvement in validation performance. These results suggest that relying solely on spatial information captured by GNNs, without incorporating temporal dynamics, may be insufficient for effectively addressing the seizure detection task.

2) LSTM-GNN: Among the three models described previously, the Bi-LSTM combined with GATs demonstrated the best performance, achieving a best test F1 score of 0.82. While GCNs fit the training data well, they struggled a bit to generalize, resulting in a lower test F1 of 0.77. The Graph Transformer variant performed the worst, with a maximum test F1 of 0.692, likely due to the limited dataset size, which constrained the model's ability to leverage its high capacity effectively. All models consistently performed better when using frequency-domain inputs compared to time-domain data.

When trained on time-domain data, models tended to overfit within just a few epochs and failed to generalize effectively. Additionally, models trained on unnormalized data generalized better than those trained on normalized signals, possibly because normalization removed important amplitude variations in EEG signals that are critical for distinguishing seizure patterns. Performance also improved when using inverse distance as edge weights instead of no weights, suggesting that incorporating spatial relationships between EEG channels helps the models better capture relevant brain activity patterns.

3) GNN-LSTM: The 1 Layer GCN+LSTM performed the best among the four variants followed by 1 Layer GAT+LSTM. The GAT variants are not given the Distance Adjacency Matrix which may be a reason that they perform slightly less than the GCN variants. The 2 Layer variants perform worse with a test MacroF1 of 0.609 for the 2 Layer GCN+LSTM variant, which suggests oversquashing may be happening.

4) STGCN: Both models [Adaptive Avg Pooling, Attention pooling] were trained for 50 epochs using a learning rate of 5×10^{-4}, a weight decay of 1×10^{-5}, and gradient clipping with a maximum norm of 1.0. Data augmentation was also applied during training, which empirically improved stability of the training process for this model.

On the robust validation set, the first model (with average pooling) achieved an F1 score of 0.82, while the *attention* based pooling using softmax on the STGCN model achieved an F1 of 0.80 on the same robust validation set. Based on these results, the simpler average pooling model was selected for testing which yielded a score on test set of 0.76. Additionally, it was observed that both variants performed better when trained on time-domain data compared to frequency-domain representations. A plausible explanation is that the STGCN architecture already incorporates a spectral decomposition through the graph Laplacian, implicitly introducing a Fourier-like transformation.

5) S4-GNN: Two hybrid models S4-GCN and S4-GAT were trained which apply a single S4 layer to each EEG channel independently. The resulting per-channel embeddings are then passed to a graph neural network—two GCN layers or three GAT layers respectively. Models are trained using the Adam optimizer for 200 epochs with OneCycleLR scheduling (max LR 0.001), weight decay of 10^{-5}, and gradient clipping (max norm 1.0). The GAT variant uses 4 attention heads and residual connections, while the GCN variant uses graph normalization and residual projections.

Ablation experiments showed that using a patient-separated validation set was essential for reliable evaluation and training without patient separation led to overfitting and overestimated performance. Time-domain inputs outperformed frequency-domain (FFT-based) inputs by a small margin.

6) DCRNN: Both the variants of DCRNN were trained using Adam Optimizer with two diffusion steps and two stacked DCGRU layers. The 'tanh' activation function is used in DCGRU layers. The best result for both models was obtained at dropout 0 and 1×10^{-3} learning rate with cosine annealing schedule. The models did not overfit with validation F1 (0.7174 for distance and 0.6401 for correlation) stabilizing around 100 epochs. The correlation graph's adjacency matrix is computed on the fly for each input batch to capture the evolving brain dynamics (Test F1 score 0.75721) which may have led it to perform marginally better than the distance graph variant (Test F1 score 0.74920).

C. Combined Results

Table I shows the best results obtained from various models for comparison. From the table, S4 + GCN stands out as the top performer, closely followed by LSTM + GAT. S4 proves to be the strongest temporal model among those tested. The superior performance of S4 + GCN likely stems from S4's ability to capture long-range temporal dependencies combined with GCN's strength in modeling spatial relationships, which is often more robust for structured spatial data like EEG than dynamic attention mechanisms like GATs. Furthermore, frequency-domain inputs led to better performance in models such as LSTM, LSTM-GNN variants, and DCRNN, whereas time-domain data yielded better results for S4-GNN variants and STGCN.

TABLE I
BEST RESULTS BY MODEL AND DATA TYPE

Model	Test F1 (Time data)	Test F1 (Freq. data)
S4-GCN	**0.83765**	0.83631
LSTM-GAT	0.73549	**0.82047**
S4	—	**0.81459**
Resnet-LSTM	**0.81352**	0.49908
S4-GAT	**0.80212**	0.80066
LSTM-GCN	0.69459	**0.77297**
STGCN [Avg pooling]	**0.76001**	0.67699
Corr-DCRNN	—	0.75721
Dist-DCRNN	—	0.74920
GCN-LSTM	0.64124	**0.74132**
GAT-LSTM	0.67059	**0.74101**
LSTM	0.69459	**0.72943**
GAT	—	0.56997

IV. CONCLUSION

This study demonstrates the effectiveness of incorporating spatial information through graph-based models for EEG-based seizure detection. While temporal models like S4 already achieve strong performance (F1 = 0.814), hybrid approaches that integrate spatial and temporal representations further improve the test score, most notably, the S4-GCN model achieves an F1 score of 0.838. Graph-only models like GNNs underperform in isolation (F1 = 0.569), underscoring the importance of temporal modeling. These findings highlight that leveraging both spatial and long-range temporal structure is critical for robust seizure detection and suggest that graph-enhanced models offer a promising direction for improving clinical EEG analysis.

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